Individual Report

AIB Microalgae Commercialization Project

Group 21

Supervisor: Dr. Yap Yee Jiun

Date: April 29th, 2026

Student Name	School of CS OWA
Rajul Kabir	hcyrk2

Other Group Members:

- Shoeb Mirza bin Mohammed Faizal
- Nikitha Rajagopalan
- Muhammad Umar

Table of Contents

1. Environment Overview

LightPhotobioreactorEnv (in **light_env.py**) is a streamlined Individual-Based Model (IBM) photobioreactor simulation focused on **light-efficient Spirulina cultivation**. This variant tracks cells in **1D (vertical depth only)**, uses a simplified single-spectrum light model, and operates without a formal difficulty system. It is designed to be computationally efficient while retaining the core biological physics needed for light and nutrient management research [7].

A key design distinction is the **deliberate absence of a grace period**. The agent has full actuator authority from step 0, making early-episode stabilisation harder and demanding more careful initial control.

Parameter	Value
Reactor volume	30 L
Reactor depth	30 cm
Max cells	300,000
Default initial cells	2,000
Simulation time step (dt)	0.01 h
Episode length	14,400 steps (≈ 144 h / 6 days)
Spatial tracking	1D (z-axis depth only)

2. Action & Observation Space

The agent controls four continuous actuators normalised to [-1, 1], decoded via Exponential Moving Average (EMA) filters. An ActionSmoothnessWrapper applies an L2 penalty on consecutive action deltas (coefficient = 0.001) to suppress chattering; this is subtracted from the step reward.

2.1 Action Space

Index	Actuator	Decoded Range	EMA Coefficient
0	Stirring speed	50 – 200 RPM	$\alpha = 0.05$
1	Surface light intensity	0 – 2000 μmol/m²/s	Instant
2	Nutrient flow rate	0 – 100 mg/h	$\alpha = 0.06$
3	CO_2 injection rate	0 – 120 mL/min	$\alpha = 0.15$

The CO_2 range is broadened to 0–120 mL/min. This design change prevents the agent from "reward hacking" the pH mechanics by relying on its default Gaussian initialization (which previously mapped exactly to the optimal 2.5 mL/min biological requirement), thereby forcing it to actively learn pH regulation.

2.2 Observation Space

The agent receives a 6-dimensional sensor vector. Dissolved O_2 and CO_2 are tracked as full physics states but are **not included in the observation** — the agent must infer their status from indirect proxy sensors. True optical density (OD) is similarly unobserved; turbidity (NTU) serves as its noisy proxy. Observations are subject to step-level jitter drawn from U(0.98, 1.02) and a per-episode static calibration drift drawn from U(0.98, 1.02). The pH channel additionally receives RPM-coupled EMA smoothing (2–6 step lag). VecNormalize clips all values at clip_obs = 100.0.

Index	Sensor	Units
0	Turbidity (NTU)	0 – 5000
1	pH	0 – 14
2	Nutrient concentration	0 – 5000 mg/L
3	Temperature	0 – 50 °C
4	Conductivity	0 – 10000 µS/cm
5	RGB absorbance	0 – 20

3. Physics Model

3.1 Strain Randomisation

At each episode reset, a unique algal strain is drawn from biologically-grounded Gaussian priors. Randomisation is unconditional — there is no difficulty gate:

Parameter	Mean	Std / Range	Description
mu_max	0.10 h^{-1}	0.05	Maximum specific growth rate
Ks	20.0 mg/L	5.0	Monod nutrient half-saturation
Ki	120.0 µmol/m²/s	30.0	Photoinhibition constant
T_opt	27.0 °C	2.0	Optimal growth temperature
Q_min	1.5	—	Droop minimum quota [5]
tau_acclim	—	U(1, 4) h	Photo-acclimation time constant

Additionally, mu_max and Ks wander by ±1% every 500 steps throughout every episode, requiring the policy to adapt continuously to a slowly shifting strain.

3.2 Growth Model

Net growth uses a simplified single-spectrum photoinhibition model without explicit CO_2 toxicity (f_CO2_tox) or cumulative membrane fatigue (fatigue_tax) terms [4]:

```
current_mu = (mu_max * f_I * f_Q * f_temp * f_shock
              * f_O2 * f_pH * f_Osmosis * f_CO2 * repair_tax)
net_mu = current_mu - m_respiration
```

The five primary factors are listed below; the full 10-factor table is in Appendix A.1.

Factor	Description
f_I	Light saturation + photoinhibition
f_Q	Droop internal nutrient quota [5]
f_temp	Gaussian temperature factor ($\sigma = 5$ °C)
f_O2	Dissolved O_2 toxicity (threshold 26 mg/L)
f_pH	Asymmetric Gaussian pH inhibition (peak 9.5)

A **fouling factor** accumulates proportionally to culture density every 100 steps and exponentially attenuates the surface irradiance received by the culture. Under high irradiance (>1000 μmol m^{-2} s^{-1}) or nutrient stress (<100 mg/L), the model **bleaches** Spirulina's phycocyanin pigment, reducing per-cell light absorption until conditions recover.

3.3 Cell Transport (1D)

Mixing uses sinusoidal macro- and micro-turbulence along the z-axis, avoiding horizontal convection to minimise computational overhead. At low mixing intensities, cells sediment via Fickian diffusion (passive downward drift under gravity). However, at higher RPMs, a **turbulent flash-light effect** accurately replicates the biological Kok effect: stochastic mixing rapidly cycles cells between the heavily irradiated photic zone at the surface and the dark rest zone at the bottom, increasing overall photosynthetic efficiency.

Crucially, light attenuation features full **RGB spectral splitting**. Surface irradiance is divided into red (40%), blue (40%), and green (20%) bands, each attenuating at different extinction coefficients down the z-axis. Growth is driven exclusively by red PAR (Photosynthetically Active Radiation), but total PAR triggers photoinhibition. The extinction coefficients also dynamically scale with turbidity, bubble scattering (from RPM), and cell clumping. The full transport equations are in Appendix A.2.

3.4 Flocculation and Probabilistic Lysis

Flocculation follows mean-field Smoluchowski dynamics [6] (a collision-rate model for how particles clump) each step: sticking probability rises with optical density (OD) and drops under high-RPM shear, while Brownian breakup limits runaway aggregation when mixing is low. Clumps are particularly costly because they self-shade ($f\_clump\_shade = M^{-1/3}$), cutting light access for the cells trapped inside them.

Rather than a hard death threshold, starvation is modelled via a **probabilistic lysis model**. Background die-off runs at ~0.5% per day, but as internal quotas (f_Q) fall, that rate climbs to ~5%. Getting the nutrient feed wrong even briefly can compound into a die-off that is very difficult to recover from across a 6-day episode.

3.5 Gas Exchange & pH

The environment uses mass-balance for gas exchange. Dissolved Oxygen (DO_2) accumulates via photosynthesis (~1.2 mg O_2 per mg biomass) and drains through gas transfer governed by $k_L a$ (the volumetric mass transfer coefficient). Without a DO_2 stripping actuator, the agent must use stirring to prevent DO_2 from exceeding the 26 mg/L toxicity threshold. Dissolved inorganic carbon (DIC) evolves via equilibration, direct injection, and consumption (~0.015 mg CO_2 per mg biomass).

pH dynamics, the core proxy reward signal, are modeled as a blend of carbonate chemistry and biotic drift. This state is filtered via a 95% EMA to simulate industrial probe lag. The system targets a Zarrouk-like buffer of pH 9.5 [10] — the standard alkaline growth medium for Spirulina. $k_L a$ scales with stirring but is damped by viscosity changes from high OD or clumping.

Temperature is not static either: the impeller adds viscous heat scaling with RPM^3 (up to ~0.5°C at 200 RPM), surface light contributes additional warming, and the reactor cools passively toward 25°C ambient. Since conductivity is temperature-scaled, stirring speed also indirectly shapes that observed signal.

4. Reward Function

The reward is focused on **mass productivity and oxygen dynamics** using proxy signals tuned to the simplified physics model:

```
reward = (productivity * 1e-8) + (n_spawns * 0.1) + (mean_f_Q * 0.05) \
       + reward_do2 + reward_growth_rate \
       - penalty_do2 - penalty_clump - action_smooth_penalty
```

Stagnation and crash penalties are conditional/terminal and omitted from the inline formula above; they appear below and in the full table in Appendix A.2.

Component	Signal	Purpose
productivity × 1e−8	Net mass change per step	Primary biomass signal
n_spawns × 0.1	Count of division events	Rewards successful division
mean_f_Q × 0.05	Mean internal quota	Nutrient adequacy
reward_growth_rate	max(0, dOD/dt) × 20 × pop_factor	Dense derivative growth signal
penalty_do2	$-0.002 \times \max(0, DO_2-18)^2$	O_2 toxicity deterrent
reward_stagnation	−0.05 if OD High Water Mark (HWM) stale > 800 steps	Anti-plateau penalty
penalty_crash	−1000 (terminal)	Population collapse deterrent

The reward function is designed to encourage sustained, safe productivity while discouraging risky or degenerate behaviours. Early tuning revealed policies that occasionally exploited single proxies during transient physics regimes; to counter this the final reward mixes direct productivity, division counts and anti-stagnation penalties so that short-lived proxy gains do not yield sustained reward. Coefficients were adjusted through short calibration runs and trace inspection to reduce proxy exploitation.

The full 10-component table is in Appendix A.2. A ***population factor*** *(pop_factor = 1 + max(0, 1 − num_active/6000)) amplifies growth rewards as population falls, incentivizing recovery from die-off events. A population below 10 active cells triggers immediate termination.*

5. Algorithm Selection: Recurrent PPO

5.1 Why Recurrent PPO

LightPhotobioreactorEnv is **partially observable by design** [9]. Dissolved O_2, dissolved CO_2, and true cell mass are all hidden from the observation vector—intentionally, to push the agent toward strategies that would generalise to real reactors where these states are rarely measured directly.

Standard PPO [1] is Markovian: it maps each sensor snapshot to an action with no memory of what came before. That breaks down here. A memoryless agent cannot tell whether a rising pH reflects genuine photosynthetic carbon drawdown or just valve lag from a recent CO_2 injection—two situations that call for opposite responses.

RecurrentPPO [3] handles this by pairing the actor-critic with an LSTM [2]. The hidden state h_t builds up a running picture of the culture's trajectory across the episode, letting the policy track quantities it cannot observe directly. In practice this means it can preemptively ramp CO_2 ahead of a pH spike rather than reacting after the fact.

5.2 Hyperparameters & Entropy

```
model = RecurrentPPO(
    "MlpLstmPolicy", vec_env,
    learning_rate=3e-4, n_steps=7200,
    batch_size=256,     n_epochs=4,
    ent_coef=ENTROPY_INIT, gamma=0.995,
    policy_kwargs={"lstm_hidden_size": 256, "n_lstm_layers": 1},
)
```

The full hyperparameter table is in Appendix A.3. The entropy coefficient follows a hybrid decay: an exponential baseline modulated by a std-band controller that boosts exploration if policy std < 0.20 and reduces it if std > 0.45 [8], keeping the policy in a productive exploration band throughout training.

6. Training Curriculum

LightPhotobioreactorEnv is fully compatible with the CurriculumStartWrapper and PopulationStitchCallback infrastructure from recurrent_ppo.py and can be used as a warm-up or standalone training environment. When used with recurrent_ppo.py, the curriculum varies initial population distributions and gates harder start modes using mastery thresholds computed over 100,000-step chunks. In standalone mode, as described in Section 6.2, no such gates apply.

6.1 Episode Start Modes

Three population initialisation modes are sampled per-episode. **Low** starts draw a log-uniform 1,000–4,000 cell population. **Mid** starts initialise at 6,000–10,000 cells to challenge mid-growth management. **Stitched** starts restore a full physical state snapshot from a previously successful high-population episode. Because light_env is 1D, cells_x is not stored or restored. Stitched starts are gated until 20 completed episodes. The full START_DISTRIBUTION dictionary is in Appendix A.4.

6.2 Advancement and Evaluation

There is no formal difficulty curriculum here—no D0/D1/D2 stages or mastery streaks. Every episode uses the same physics, and progress is judged by how well the policy maintains stability and avoids crashes across varied starting populations. Growth factor (peak_OD / init_OD_equivalent) is still logged but does not gate anything. The 2D predecessor to this environment did implement a staged curriculum, but the PPO struggled to make sense of the added spatial complexity, so that design was scrapped in favour of this simpler 1D formulation.

6.3 Infrastructure

Five callbacks manage training infrastructure: checkpointing, episode metrics collection, warm-start state saving, live TQDM display, and TensorBoard entropy logging. Full descriptions are in Appendix A.5. The entropy coefficient and policy standard deviation are also written to TensorBoard each rollout, which proved useful for catching policy collapse early during training runs.

7. Evaluation & Results

I evaluated the model externally by benchmarking the trained RecurrentPPO agent against a smoothed RandomAgent baseline. The evaluation used a 30,000-step horizon—exceeding the 14,400-step training episode length—to test long-horizon generalisation beyond the trained episode boundary. Normalisation statistics were loaded and applied to maintain consistency with the training distribution.

3 Episodes × 30,000 Steps · Light Env · Standard Configuration

Metric	RecurrentPPO	Random	Δ (absolute)
Total Reward	12411.20	822.79	+11588.41
Peak OD	0.0917	0.0074	+0.0843
Final Population	66452	5347.67	+61104.33
Crash Rate	0%	0%	—
Avg CO_2 Injection (mL/min)	56.33	60.45	-4.12

The RecurrentPPO achieved roughly 15× the cumulative reward and 12× the final population of the random baseline while using slightly less CO_2 on average (56.33 vs 60.45 mL/min), suggesting more consistent pH regulation (and well timed injections of CO_2). Both agents avoided outright crashes, but the random baseline plateaued at low density while the PPO sustained growth across the full 30,000-step window.

8. Reflection

Developing LightPhotobioreactorEnv highlighted the trade-off between biological fidelity and training stability. A critical design choice was the removal of the "grace period," granting the agent full authority from step 0. This forced the agent to handle early-episode instability—often exacerbated by aggressive CO_2 and inconsistent lighting, necessitating the use of stitched episodes to maintain a stable and high-density reward signal.

The 1D vertical model prioritises computational speed, allowing for rapid iteration on depth-dependent light and spectral attenuation. While my decision to omit horizontal convection and airlift heterogeneity limits its use as a full "digital twin," it serves as an effective tool for exploring lighting optimisation and lysis risks.

Reward calibration was simplified by using fewer proxy signals. However, the stagnation penalty sometimes conflicted with growth rewards in physics-constrained low-light regimes. Interestingly, the agent often discovered a constant CO_2 injection strategy to stabilise pH, a behaviour that emerged with and without CO_2 toxicity.

I also learnt by trial and error, the difficulties of working with entropy control, requiring extensive testing and tuning to balance exploration and exploitation. In the end, I opted for an approach that still encouraged exploration early in the training while maintaining stability later on.

Finally, I highlight that the recurrent architecture remains essential. By integrating temporal sensor data, the agent filters noise and manages mixing-induced lags. This memory allows the policy to track hidden biological states, like dissolved oxygen accumulation or pH shifts, confirming that recurrent policies are vital for controlling biological systems where state variables are often unobserved.

9. Limitations & Future Work

LightPhotobioreactorEnv is a streamlined testbed with several known gaps. The 1D vertical model eliminates horizontal heterogeneity, so the agent never encounters the spatial gradients that real airlift or paddle-wheel reactors produce. The simulation is also single-species: competitive dynamics, contamination, and predator-prey interactions are absent. Sensor noise is limited to multiplicative jitter and static calibration drift; structured faults like probe fouling or actuator saturation are not modelled.

Future work should focus on validating policies on physical reactors and extending the model to multi-species scenarios. Thermodynamically grounded gas exchange and alternative training curricula—or model-based approaches that explicitly estimate hidden states—could also improve sample efficiency and robustness.

References

[1] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347*.

[2] Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.

[3] Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., & Dormann, N. (2021). Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research*, 22(268), 1–8.

[4] Andrews, J. F. (1968). A mathematical model for the continuous culture of microorganisms utilizing inhibitory substrates. *Biotechnology and Bioengineering*, 10(6), 707–723.

[5] Droop, M. R. (1968). Vitamin B12 and marine ecology. IV. The kinetics of uptake, growth and inhibition in Monas lutheri. *Journal of the Marine Biological Association of the United Kingdom*, 48(3), 689–733.

[6] von Smoluchowski, M. (1917). Versuch einer mathematischen Theorie der Koagulationskinetik kolloider Lösungen. *Zeitschrift für physikalische Chemie*, 92, 129–168.

[7] Richmond, A. (Ed.). (2004). *Handbook of Microalgal Culture: Biotechnology and Applied Phycology*. Blackwell Science.

[8] Ahmed, Z., Roux, N. L., Norouzi, M., & Schuurmans, D. (2019). Understanding the Impact of Entropy on Policy Optimization. *Proceedings of the 36th International Conference on Machine Learning (ICML 2019)*, PMLR 97.

[9] Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

[10] Zarrouk, C. (1966). *Contribution à l'étude d'une cyanophycée: Influence de divers facteurs physiques et chimiques sur la croissance et la photosynthèse de Spirulina maxima*. PhD thesis, Université de Paris.

Appendix A — Full Technical Details

A.1 Full Growth Factor Table

Factor	Description	Key Parameters
f_I	Light saturation + photoinhibition	Monod with photoinhibition: $Ki + I^2/2500$
f_Q	Droop internal quota	Q_min = 1.5
f_temp	Gaussian temperature factor	σ = 5 °C around T_opt
f_shock	Photo-acclimation mismatch penalty	Coefficient 3×10^{-6}
f_O2	Dissolved O_2 toxicity	Threshold 26 mg/L
f_pH	Asymmetric Gaussian pH inhibition	Peak 9.5; σ_acid=1.2, σ_base=2.0
f_osmosis	Osmotic stress	Gaussian penalty above 2000 mg/L nutrients
f_CO2	Carbon limitation	Kc = 0.5 mg/L
repair_tax	Shear repair cost	Onset 150 RPM; max 25% penalty at 200 RPM
m_respiration	Maintenance cost (subtracted from net_mu, not multiplicative)	1.0% of mu_max per hour

A.2 Cell Transport Equations & Full Reward Table

1D Transport:

```
v_macro = 0.005 · mix_intensity · sin(100·z − 5·t)
v_micro = 0.002 · mix_intensity · sin(500·z − 20·t + cell_index)
dz = (v_macro + v_micro) · dt_sec + √(2D) · ξ
```

Full Reward Components:

Component	Signal	Description
productivity × 1e−8	Net mass change per step	Primary biomass signal
n_spawns × 0.1	Count of division events	Rewards successful division
mean_f_Q × 0.05	Mean internal quota	Nutrient adequacy
reward_do2	O_2 production × 0.05	Photosynthesis proxy
reward_growth_rate	max(0, dOD/dt) × 20 × pop_factor	Derivative growth signal
penalty_do2	$-0.002 \times \max(0, DO_2-18)^2$	O_2 toxicity deterrent

Component	Signal	Description
penalty_clump	−0.01 × (mean_clump−1)	Anti-aggregation signal
reward_stagnation	−0.05 if OD High Water Mark (HWM) stale > 800 steps	Anti-plateau penalty
action_smooth_penalty	$-\text{coef} \times \Sigma(\Delta a)^2$	Actuator smoothness
crash penalty	−1000 (terminal)	Crash deterrent

A.3 Full Hyperparameter Table

Hyperparameter	Value	Rationale
Policy	MlpLstmPolicy	LSTM for POMDP handling
LSTM hidden size	256	Sufficient for 6D → 4D control
LSTM layers	1	Single layer avoids vanishing gradients
Learning rate	3×10^{-4}	Adam default; stable for this reward scale
n_steps	7,200	≈ 30 simulated days per update batch
batch_size	256	Standard mini-batch
n_epochs	4	Conservative PPO clip
γ (gamma)	0.995	Long-horizon discounting
Entropy coef (init)	0.02	Hybrid decay
VecNormalize clip	100.0	Bounds extreme physics excursions
Total budget	40,000,000 steps	—

A.4 Episode Start Distribution

```
START_DISTRIBUTION = {
    0: {"low": 0.85, "mid": 0.10, "stitched": 0.05},   # early training
    1: {"low": 0.40, "mid": 0.40, "stitched": 0.20},   # mid training
    2: {"low": 0.275, "mid": 0.275, "stitched": 0.45}, # late training
}
```

A.5 Callback Stack

Callback	Purpose
CheckpointCallback	Saves model every 10,000 global steps
EpisodeMetricsCallback	Records growth factor, crash rate, reward std per episode
PopulationStitchCallback	Saves warm-start states when episode-end population > 11,000
TQDMActionCallback	Live display of actuator values and rolling mean OD

Callback	Purpose
EntropyLoggingCallback	Writes entropy coefficient to TensorBoard each rollout end

Appendix B — G52GRP Peer Assessment Form

Name of assessed group member: ______Shoeb Mirza bin Mohammed Faisal______________________________

	None	Lacking	Adequate	Good	Excellent
Research & information gathering					X
Creative input					X
Co-operation within group					X
Communication within group					X
Concrete contribution[1]					X
Attendance at meetings					X

Justification of assigned marks:

Shoeb worked on the Prophet model for time-series forecasting and worked on the integration of the separate components of the project. He contributed to many early sections of the report, and was always communicative and present for meetings. He was clear and always adaptable to change.

Name of assessed group member: __Muhammad Umar_________________________________

	None	Lacking	Adequate	Good	Excellent
Research & information gathering					X
Creative input					X
Co-operation within group					X
Communication within group					X
Concrete contribution[2]					X
Attendance at meetings					X

1 Quality and quantity of concrete contribution to *group deliverables*: writing, coding, testing, open day display, preparations for presentations, etc.

2 Quality and quantity of concrete contribution to *group deliverables*: writing, coding, testing, open day display, preparations for presentations, etc.

Justification of assigned marks:

Umar worked on the Frontend and was clearly able to take feedback and concretely and consistently change any recommendations. He wrote up the presentation slides for our presentation and contributed heavily to the mid sections of the report. He was punctual and clear on his contributions.

Name of assessed group member: _____Nikitha Rajagopalan_______

	None	Lacking	Adequate	Good	Excellent
Research & information gathering					X
Creative input					X
Co-operation within group					X
Communication within group					X
Concrete contribution[3]					X
Attendance at meetings					X

Justification of assigned marks:

Nikitha worked on the backend and database of our project, and was able to help connect our pipeline to the EE team's one. She was very capable throughout the project, and progressively kept our backend up to date and was quick to any changes necessary. She contributed to some of the later sections of the report.

3 Quality and quantity of concrete contribution to *group deliverables*: writing, coding, testing, open day display, preparations for presentations, etc.